

Supplementary Material

Conformal Robot Action Sets for Robot Policies under Visual Shift

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Abstract

This supplement gives the full methodological and experimental record for Conformal Robot Action Sets (CoRAS). It formalizes the set-valued robot-action interface, gives the split-conformal proof details, defines all baselines and metrics, specifies the data splits and hyperparameters, and reports extended results for controlled simulation, PushT visual shifts, DROID-100 visual shifts, and auxiliary AutoEval online-log diagnostics. The scope of the empirical claim is offline and log-based calibration of robot action sets under visual shift; no new closed-loop physical-robot deployment is claimed.

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1 Set-valued robot-action formulation

A robot observation $x \in \mathcal{X}$ may include a single image, a history of images, proprioception, and an optional language instruction. A robot scorer g_θ maps x to scores over a finite candidate set $\mathcal{Y}(x)$. The candidate set can represent discrete action tokens, skill tokens, object targets, waypoints, grasp proposals, or a finite discretization of continuous actions. In the experiments, continuous robot actions are converted into finite labels by a source-domain K -means action codebook. When the original continuous action $a \in \mathbb{R}^d$ is available, each token $y \in \{1, \dots, K\}$ has an associated center $c_y \in \mathbb{R}^d$.

Given a calibration pair (x_i, y_i) , a nonconformity score $S_\phi(x_i, y_i)$ is small when the demonstrated candidate is compatible with the observation. For the probability-score version used in the main CoRAS experiments,

$$S_\phi(x, y) = 1 - p_{\theta, \phi}(y \mid x), \quad (1)$$

where ϕ denotes a small target-domain adapter and θ is fixed. The conformal action set is

$$\Gamma_\alpha(x) = \{y \in \mathcal{Y}(x) : S_\phi(x, y) \leq \hat{q}_\alpha\}. \quad (2)$$

The set is a statistical interface to a downstream robot stack. A compact set indicates that the adapted policy assigns sufficient mass to a small group of candidates. A large set indicates that many candidate actions remain statistically plausible at the selected coverage level.

2 Split protocol and algorithmic details

The experiments use four disjoint data roles: source training, target tuning, target calibration, and target testing. The source split trains or loads the base action-token predictor. The target tuning split updates only the adapter. The target calibration split computes the conformal threshold. The target test split is used once for evaluation. This split discipline is necessary because reusing calibration examples for adapter fitting can invalidate the exchangeability argument.

2.1 CoRAS procedure

For a fixed error level α , CoRAS proceeds as follows.

1. Train or load a base scorer g_θ on source-domain robot data.
2. Freeze θ and tune a small adapter p_ϕ on $\mathcal{D}_{\text{tune}}$.
3. Compute true-label nonconformity scores $s_i = S_\phi(x_i, y_i)$ on $\mathcal{D}_{\text{cal}}$.
4. Let $n = |\mathcal{D}_{\text{cal}}|$ and set $\hat{q}_\alpha$ to the conservative empirical quantile with rank $\lceil (n+1)(1-\alpha) \rceil$.
5. Return the action set $\Gamma_\alpha(x) = \{y : S_\phi(x, y) \leq \hat{q}_\alpha\}$ at test time.

If the conformal rank exceeds n , the threshold is $+\infty$ and the prediction set is the full candidate set. This behavior is conservative and occurs only when the calibration set is too small for the selected α .

2.2 Baselines

All baselines use the same train/tune/calibration/test splits.

Top-1 singleton.

The policy outputs only the highest-probability candidate.

Calibrated top-k.

The smallest integer k is calibrated so the true label is contained in the top- k set at the target level.

Vanilla conformal.

Split conformal prediction using the base source-trained model without target adaptation.

APS conformal.

Adaptive prediction sets using cumulative probability mass as the nonconformity score.

Temperature conformal.

Temperature-scaled probabilities are calibrated and then conformalized.

Prompt top-1.

The target adapter is tuned but the output remains a singleton.

CoRAS.

Adapter tuning followed by split conformal calibration using probability-score nonconformity.

CoRAS-APS.

Adapter tuning followed by APS-style conformal calibration.

Mondrian CP.

Domain-conditional conformal thresholds, where sufficient domain labels are available.

3 Coverage results

Theorem 1 (Finite-sample marginal coverage). *Let the adapter p_ϕ be fitted without using $\mathcal{D}_{\text{cal}}$. Conditional on the fitted adapter, suppose the calibration examples $(X_i, Y_i)_{i=1}^n$ and the test example (X_{n+1}, Y_{n+1}) are exchangeable. Then*

$$\mathbb{P}\{Y_{n+1} \in \Gamma_\alpha(X_{n+1})\} \geq 1 - \alpha. \quad (3)$$

Proof. Condition on the fitted adapter p_ϕ . The true-label scores $S_\phi(X_1, Y_1), \dots, S_\phi(X_n, Y_n), S_\phi(X_{n+1}, Y_{n+1})$ are exchangeable. Therefore the rank of the test score among the $n + 1$ scores is uniform up to ties when randomized, and conservative under deterministic tie handling. The event $Y_{n+1} \notin \Gamma_\alpha(X_{n+1})$ is equivalent to the test score exceeding the selected calibration quantile. The conformal rank choice bounds this event by α . $\square$

Lemma 1 (Monotone score invariance). *Let $h : \mathbb{R} \rightarrow \mathbb{R}$ be strictly increasing. Replacing S_ϕ by $h \circ S_\phi$ leaves the conformal set unchanged.*

Proof. A strictly increasing transform preserves all pairwise score comparisons and therefore preserves both the empirical quantile rank and the membership test $S_\phi(x, y) \leq \hat{q}_\alpha$. $\square$

Proposition 1 (Distribution-shift slack). *For a fixed fitted adapter and threshold q , let $A_q = \{(x, y) : S_\phi(x, y) \leq q\}$. For calibration and deployment distributions P_{cal} and P_{test} ,*

$$P_{\text{test}}(A_q) \geq P_{\text{cal}}(A_q) - d_{\text{TV}}(P_{\text{test}}, P_{\text{cal}}). \quad (4)$$

Equivalently, if the score CDFs differ uniformly by at most Δ , the deployment coverage differs from calibration coverage by at most Δ .

Proof. The total-variation inequality $|P(A) - Q(A)| \leq d_{\text{TV}}(P, Q)$ applied to A_q gives the result. The CDF statement is the same argument applied to score-threshold events. $\square$

Proposition 2 (Planner-side decomposition). *Suppose a planner executes directly only when a gate $G(X) = 1$ accepts the conformal set, and suppose that $Y \in \Gamma_\alpha(X)$ and $G(X) = 1$ imply existence of an admissible planner action. If fallback failure probability is at most ρ_F , then*

$$\mathbb{P}(\text{execution failure}) \leq \mathbb{P}\{Y \notin \Gamma_\alpha(X)\} + \rho_F \mathbb{P}\{G(X) = 0\}. \quad (5)$$

Proof. Partition failures into failures under an accepted set and failures after fallback. Under the stated planner condition, an accepted set can fail only if the true candidate is outside the set. Fallback contributes at most ρ_F conditional on being invoked. $\square$

Table 1: Dataset and split summary used in the supplementary experiments.

Track	Frames	Episodes	K	Action dim.	Source frames	Target frames
Synthetic controlled shift	9,000	450	16	3	7,027	1,973
PushT visual shift	25,650	206	32/64/128	2	15,491	10,159
DROID-100 visual shift	32,212	100	64/128	7	19,562	12,650
AutoEval online logs	1,575	37	64	7	1,045	530

All frames are resized to the resolution shown in the configuration files. PushT and DROID source frames are clean real robot observations; target frames are generated visual-shift variants. AutoEval frames are online real-robot log frames with raw 7D action labels.

4 Datasets, domains, and configurations

The visual-shift datasets use clean real robot frames as source data and construct target domains by deterministic image transformations. The shift domains are lighting, camera crop, occlusion, and blur/noise for PushT and DROID-100. Synthetic data use camera, lighting, and occlusion target shifts generated directly by the simulator.

The main PushT and DROID runs use a ResNet-18 visual encoder, pretrained initialization, frozen encoder from the start, adapter rank 32, learning rate 3×10^{-4} for source training, adapter learning rate 10^{-3} , weight decay 10^{-4} , target tuning fraction 0.15, test fraction 0.25, and calibration fractions $\{0.10, 0.25, 0.40\}$. PushT uses batch size 128; DROID-100 uses batch size 64. Both use 8 source epochs and 5 adapter epochs. AutoEval diagnostics use the same architecture with 6 source epochs and 4 adapter epochs.

5 Metrics

The following metrics are used throughout the supplementary tables.

- Coverage.** Fraction of test frames for which the true action token is contained in the set.
- Coverage gap.** Absolute deviation from the target coverage $1 - \alpha$.
- Mean set size.** Average cardinality of $\Gamma_\alpha(x)$.
- Set/K.** Mean set size divided by the action-codebook size.
- Fail-to-abstain.** Fraction of samples for which the set is empty or fails to contain the true token when a singleton would be unsafe. For singleton baselines, this is the complement of top-1 accuracy.
- Oracle-in-set L2.** Distance from the continuous demonstrated action to the nearest codebook center contained in the prediction set.
- Action L2 gain.** Top-1 continuous-action error minus oracle-in-set continuous-action error.
- Set diameter.** Mean maximum pairwise distance among action centers contained in the set.
- Episode mean coverage.** Average coverage computed per episode.
- Episode block bootstrap.** Bootstrap interval computed over episode blocks rather than over individual frames.

Table 2: CoRAS summary at $\alpha = 0.10$ across evaluation tracks.

Track	K	Method	Coverage	$ \Delta $	Mean set	Set/K	Top-1	Fail-to-abstain
Synthetic	16	CoRAS	0.900 $\pm$ 0.019	0.013 $\pm$ 0.012	5.986 $\pm$ 1.211	0.374	0.624 $\pm$ 0.039	0.009 $\pm$ 0.008
PushT	32	CoRAS	0.908 $\pm$ 0.010	0.011 $\pm$ 0.006	13.517 $\pm$ 0.473	0.422	0.242 $\pm$ 0.012	0.000 $\pm$ 0.000
PushT	64	CoRAS	0.908 $\pm$ 0.010	0.010 $\pm$ 0.008	27.169 $\pm$ 1.029	0.425	0.155 $\pm$ 0.014	0.000 $\pm$ 0.000
PushT	128	CoRAS	0.909 $\pm$ 0.010	0.011 $\pm$ 0.007	54.285 $\pm$ 1.948	0.424	0.103 $\pm$ 0.008	0.000 $\pm$ 0.000
DROID-100	64	CoRAS	0.913 $\pm$ 0.036	0.033 $\pm$ 0.016	50.608 $\pm$ 3.360	0.791	0.060 $\pm$ 0.024	0.000 $\pm$ 0.000
DROID-100	128	CoRAS	0.900 $\pm$ 0.041	0.035 $\pm$ 0.019	98.994 $\pm$ 5.746	0.773	0.071 $\pm$ 0.018	0.000 $\pm$ 0.000

Values for PushT and DROID average over 3 seeds and 3 calibration fractions. $|\Delta|$ is absolute deviation from target coverage. Set/K is included to compare action-codebook sizes.

Table 3: Synthetic controlled-shift results at $\alpha = 0.10$.

Method	Coverage	$ \Delta $	Mean set	Set/K	Top-1	Fail-to-abstain
Top-1 singleton	0.471 $\pm$ 0.082	0.429 $\pm$ 0.082	1.000 $\pm$ 0.000	0.062	0.471 $\pm$ 0.082	0.529 $\pm$ 0.082
Prompt top-1	0.624 $\pm$ 0.039	0.276 $\pm$ 0.039	1.000 $\pm$ 0.000	0.062	0.624 $\pm$ 0.039	0.376 $\pm$ 0.039
Calibrated top-k	0.908 $\pm$ 0.032	0.025 $\pm$ 0.019	10.800 $\pm$ 2.387	0.675	0.471 $\pm$ 0.082	0.000 $\pm$ 0.000
Vanilla CP	0.897 $\pm$ 0.028	0.019 $\pm$ 0.018	8.501 $\pm$ 2.265	0.531	0.471 $\pm$ 0.082	0.008 $\pm$ 0.012
APS CP	0.918 $\pm$ 0.026	0.026 $\pm$ 0.016	9.863 $\pm$ 1.274	0.616	0.471 $\pm$ 0.082	0.000 $\pm$ 0.001
Temperature CP	0.899 $\pm$ 0.027	0.017 $\pm$ 0.019	8.931 $\pm$ 2.485	0.558	0.471 $\pm$ 0.082	0.004 $\pm$ 0.008
Mondrian CP	0.898 $\pm$ 0.022	0.019 $\pm$ 0.004	6.042 $\pm$ 1.171	0.378	0.624 $\pm$ 0.039	0.009 $\pm$ 0.006
CoRAS	0.900 $\pm$ 0.019	0.013 $\pm$ 0.012	5.986 $\pm$ 1.211	0.374	0.624 $\pm$ 0.039	0.009 $\pm$ 0.008
CoRAS-APS	0.942 $\pm$ 0.011	0.042 $\pm$ 0.011	9.068 $\pm$ 0.434	0.567	0.624 $\pm$ 0.039	0.000 $\pm$ 0.000

The synthetic suite includes target camera, lighting, and occlusion shifts. The action vocabulary has $K = 16$ tokens.

6 Aggregate results

Figure 1 summarizes the main cross-track pattern. PushT maintains near-target coverage with sets occupying approximately 42% of the codebook across $K \in \{32, 64, 128\}$. DROID-100 reaches near-target marginal coverage but requires substantially larger fractions of the action vocabulary, indicating that the candidate generator is less discriminative on the more diverse real-robot data.

7 Calibration and codebook sweeps

The simulation sweep shows the expected monotone relation between α and set size. PushT shows a stable normalized-set-size pattern across K , which suggests that the result is not an artifact of a single discretization. DROID-100 shows a different regime: calibrated coverage is achievable, but the set fraction remains high across K , indicating that larger candidate vocabularies do not automatically yield sharper operational action sets.

8 Per-domain results

The per-domain tables show that visual-shift type affects both coverage and set size. On PushT $K=64$, CoRAS has lower set size than vanilla conformal prediction across all four target shifts, with the largest contrast on camera crop. On DROID-100 $K=64$, per-domain coverage remains close to the marginal target but set sizes are consistently large.

9 Continuous-action geometry

The action-geometry metrics evaluate whether a token set contains continuous actions close to the demonstrated action. These metrics are diagnostic because the action set is ultimately consumed by a robot planner in continuous action space. On PushT, CoRAS reduces top-1 action error substantially while using smaller sets

Table 4: PushT K=32 visual-shift results at $\alpha = 0.10$.

Method	Coverage	$ \Delta $	Mean set	Set/K	Top-1	Fail-to-abstain
Top-1 singleton	0.145 ± 0.010	0.755 ± 0.010	1.000 ± 0.000	0.031	0.145 ± 0.010	0.855 ± 0.010
Prompt top-1	0.242 ± 0.012	0.658 ± 0.012	1.000 ± 0.000	0.031	0.242 ± 0.012	0.758 ± 0.012
Calibrated top-k	0.916 ± 0.010	0.016 ± 0.010	22.556 ± 0.726	0.705	0.145 ± 0.010	0.000 ± 0.000
Vanilla CP	0.914 ± 0.008	0.014 ± 0.008	20.639 ± 0.440	0.645	0.145 ± 0.010	0.000 ± 0.000
APS CP	0.913 ± 0.009	0.013 ± 0.009	20.602 ± 0.570	0.644	0.145 ± 0.010	0.000 ± 0.000
Temperature CP	0.913 ± 0.011	0.015 ± 0.007	20.774 ± 0.423	0.649	0.145 ± 0.010	0.000 ± 0.000
Mondrian CP	0.900 ± 0.016	0.011 ± 0.010	12.942 ± 0.775	0.404	0.242 ± 0.012	0.000 ± 0.000
CoRAS	0.908 ± 0.010	0.011 ± 0.006	13.517 ± 0.473	0.422	0.242 ± 0.012	0.000 ± 0.000
CoRAS-APS	0.906 ± 0.008	0.009 ± 0.004	13.714 ± 0.579	0.429	0.242 ± 0.012	0.000 ± 0.000

Reported at $\alpha = 0.10$. Values are mean $\pm$ standard deviation across the experiment matrix.

Table 5: PushT K=64 visual-shift results at $\alpha = 0.10$.

Method	Coverage	$ \Delta $	Mean set	Set/K	Top-1	Fail-to-abstain
Top-1 singleton	0.089 ± 0.008	0.811 ± 0.008	1.000 ± 0.000	0.016	0.089 ± 0.008	0.911 ± 0.008
Prompt top-1	0.155 ± 0.014	0.745 ± 0.014	1.000 ± 0.000	0.016	0.155 ± 0.014	0.845 ± 0.014
Calibrated top-k	0.913 ± 0.016	0.018 ± 0.008	44.778 ± 2.224	0.700	0.089 ± 0.008	0.000 ± 0.000
Vanilla CP	0.910 ± 0.016	0.016 ± 0.008	42.036 ± 2.137	0.657	0.089 ± 0.008	0.000 ± 0.000
APS CP	0.912 ± 0.016	0.017 ± 0.008	42.299 ± 2.075	0.661	0.089 ± 0.008	0.000 ± 0.000
Temperature CP	0.911 ± 0.015	0.017 ± 0.008	42.371 ± 2.117	0.662	0.089 ± 0.008	0.000 ± 0.000
Mondrian CP	0.899 ± 0.021	0.015 ± 0.014	26.298 ± 1.821	0.411	0.155 ± 0.014	0.000 ± 0.000
CoRAS	0.908 ± 0.010	0.010 ± 0.008	27.169 ± 1.029	0.425	0.155 ± 0.014	0.000 ± 0.000
CoRAS-APS	0.909 ± 0.010	0.011 ± 0.008	27.308 ± 1.232	0.427	0.155 ± 0.014	0.000 ± 0.000

Reported at $\alpha = 0.10$. Values are mean $\pm$ standard deviation across the experiment matrix.

than vanilla conformal prediction. On DROID-100, oracle-in-set distances improve over singleton prediction, but the sets are large, reinforcing the candidate-generator bottleneck identified by the cardinality metrics.

10 Episode-level diagnostics

Frame-level conformal coverage can overstate reliability when consecutive frames are highly correlated. The episode/block diagnostics reduce this effect by aggregating over episodes. The block intervals are wider than naive frame-level summaries and should be interpreted as a more conservative measure of temporal stability.

11 AutoEval online-log diagnostics

The AutoEval logs demonstrate that the pipeline can parse online real-robot evaluation trajectories with raw action labels. The K=64 export contains 1,575 frames, 37 episodes, 24 evaluation IDs, and four language-command tasks, with all action sources recorded as raw action labels. The target-late split contains 530 frames. The failure-rate distribution is highly imbalanced, with a success mean of approximately 0.09. Accordingly, set-size failure prediction is reported only as a diagnostic. The frame-level AUCs fluctuate around chance across small target subsets, and episode-level AUCs are unstable because each diagnostic split contains only five to eight target episodes.

12 Implementation and reproducibility details

All experiments use local NumPy archives after export. This separates the calibration and evaluation code from video-decoding and remote-dataset availability. The robot-data exporters convert LeRobot-compatible datasets into a common schema with images, actions, labels, domains, episodes, and tasks. Action-token labels are generated from source-domain K -means codebooks. For PushT, the continuous action dimension

Table 6: PushT K=128 visual-shift results at $\alpha = 0.10$.

Method	Coverage	$ \Delta $	Mean set	Set/K	Top-1	Fail-to-abstain
Top-1 singleton	0.053 ± 0.007	0.847 ± 0.007	1.000 ± 0.000	0.008	0.053 ± 0.007	0.947 ± 0.007
Prompt top-1	0.103 ± 0.008	0.797 ± 0.008	1.000 ± 0.000	0.008	0.103 ± 0.008	0.897 ± 0.008
Calibrated top-k	0.914 ± 0.010	0.014 ± 0.010	90.556 ± 2.351	0.707	0.053 ± 0.007	0.000 ± 0.000
Vanilla CP	0.913 ± 0.011	0.014 ± 0.009	84.113 ± 3.336	0.657	0.053 ± 0.007	0.000 ± 0.000
APS CP	0.912 ± 0.009	0.013 ± 0.007	84.293 ± 3.307	0.659	0.053 ± 0.007	0.000 ± 0.000
Temperature CP	0.913 ± 0.008	0.013 ± 0.008	84.965 ± 2.765	0.664	0.053 ± 0.007	0.000 ± 0.000
Mondrian CP	0.906 ± 0.015	0.012 ± 0.010	54.027 ± 3.119	0.422	0.103 ± 0.008	0.000 ± 0.000
CoRAS	0.909 ± 0.010	0.011 ± 0.007	54.285 ± 1.948	0.424	0.103 ± 0.008	0.000 ± 0.000
CoRAS-APS	0.909 ± 0.009	0.010 ± 0.007	54.629 ± 2.021	0.427	0.103 ± 0.008	0.000 ± 0.000

Reported at $\alpha = 0.10$. Values are mean $\pm$ standard deviation across the experiment matrix.

Table 7: DROID-100 K=64 visual-shift results at $\alpha = 0.10$.

Method	Coverage	$ \Delta $	Mean set	Set/K	Top-1	Fail-to-abstain
Top-1 singleton	0.072 ± 0.020	0.828 ± 0.020	1.000 ± 0.000	0.016	0.072 ± 0.020	0.928 ± 0.020
Prompt top-1	0.060 ± 0.024	0.840 ± 0.024	1.000 ± 0.000	0.016	0.060 ± 0.024	0.940 ± 0.024
Calibrated top-k	0.901 ± 0.029	0.026 ± 0.011	48.667 ± 2.345	0.760	0.072 ± 0.020	0.000 ± 0.000
Vanilla CP	0.902 ± 0.043	0.035 ± 0.022	49.533 ± 3.400	0.774	0.072 ± 0.020	0.000 ± 0.000
APS CP	0.899 ± 0.042	0.034 ± 0.021	48.989 ± 3.173	0.765	0.072 ± 0.020	0.000 ± 0.000
Temperature CP	0.901 ± 0.036	0.030 ± 0.016	48.700 ± 2.784	0.761	0.072 ± 0.020	0.000 ± 0.000
Mondrian CP	0.906 ± 0.042	0.034 ± 0.021	50.483 ± 3.987	0.789	0.060 ± 0.024	0.000 ± 0.000
CoRAS	0.913 ± 0.036	0.033 ± 0.016	50.608 ± 3.360	0.791	0.060 ± 0.024	0.000 ± 0.000
CoRAS-APS	0.912 ± 0.034	0.031 ± 0.015	50.508 ± 3.061	0.789	0.060 ± 0.024	0.000 ± 0.000

Reported at $\alpha = 0.10$. Values are mean $\pm$ standard deviation across the experiment matrix.

is 2; for DROID-100 and AutoEval it is 7. The visual-shift datasets preserve original action labels while applying image-domain transformations to create target domains.

The implementation records the full result matrix in per-run directories containing split summaries, training history, calibration/evaluation metrics, prediction sets, continuous-action geometry, and episode metrics. Aggregate tables in this supplement are generated from those per-run CSV files. Model checkpoint files are not required to reproduce the tables when prediction-set CSVs and aggregate files are available.

13 Scope and assumptions

The empirical scope is offline and log-based. The results establish that CoRAS can construct calibrated action-token sets under controlled and real-frame visual shifts. They do not establish closed-loop physical-robot safety, collision avoidance, hardware reliability, or success improvements from online intervention. Those properties require an executable policy, a planner/fallback implementation, and controlled live-robot trials.

The theoretical coverage statement is marginal over the exchangeable calibration/test unit. It does not imply conditional coverage for every task, object, scene, shift type, or episode. Per-domain tables and block-bootstrap diagnostics are therefore reported as empirical stress tests rather than as additional formal guarantees. Large conformal sets are interpreted as meaningful outputs: they indicate that the calibrated interface cannot isolate a small set of plausible candidates at the chosen coverage level.

14 Summary of supplementary findings

Across all tracks, conformal calibration restores marginal action-token coverage relative to brittle singleton predictions. The most operationally favorable setting is PushT visual shift, where CoRAS maintains near-target coverage while reducing mean set size by approximately one third relative to vanilla conformal prediction across $K = 32, 64, 128$. Synthetic controlled shifts show the expected coverage-set-size tradeoff and provide a controlled validation of the method. DROID-100 shows that calibration remains feasible on a larger real-robot

Table 8: DROID-100 K=128 visual-shift results at $\alpha = 0.10$.

Method	Coverage	$ \Delta $	Mean set	Set/K	Top-1	Fail-to-abstain
Top-1 singleton	0.064 ± 0.020	0.836 ± 0.020	1.000 ± 0.000	0.008	0.064 ± 0.020	0.936 ± 0.020
Prompt top-1	0.071 ± 0.018	0.829 ± 0.018	1.000 ± 0.000	0.008	0.071 ± 0.018	0.929 ± 0.018
Calibrated top-k	0.909 ± 0.032	0.025 ± 0.019	100.778 ± 4.842	0.787	0.064 ± 0.020	0.000 ± 0.000
Vanilla CP	0.905 ± 0.041	0.034 ± 0.021	100.393 ± 6.545	0.784	0.064 ± 0.020	0.000 ± 0.000
APS CP	0.904 ± 0.038	0.029 ± 0.021	99.903 ± 5.520	0.780	0.064 ± 0.020	0.000 ± 0.000
Temperature CP	0.905 ± 0.036	0.027 ± 0.023	99.834 ± 4.872	0.780	0.064 ± 0.020	0.000 ± 0.000
Mondrian CP	0.879 ± 0.054	0.050 ± 0.025	94.873 ± 11.597	0.741	0.071 ± 0.018	0.000 ± 0.000
CoRAS	0.900 ± 0.041	0.035 ± 0.019	98.994 ± 5.746	0.773	0.071 ± 0.018	0.000 ± 0.000
CoRAS-APS	0.898 ± 0.043	0.036 ± 0.019	98.406 ± 5.839	0.769	0.071 ± 0.018	0.000 ± 0.000

Reported at $\alpha = 0.10$. Values are mean $\pm$ standard deviation across the experiment matrix.

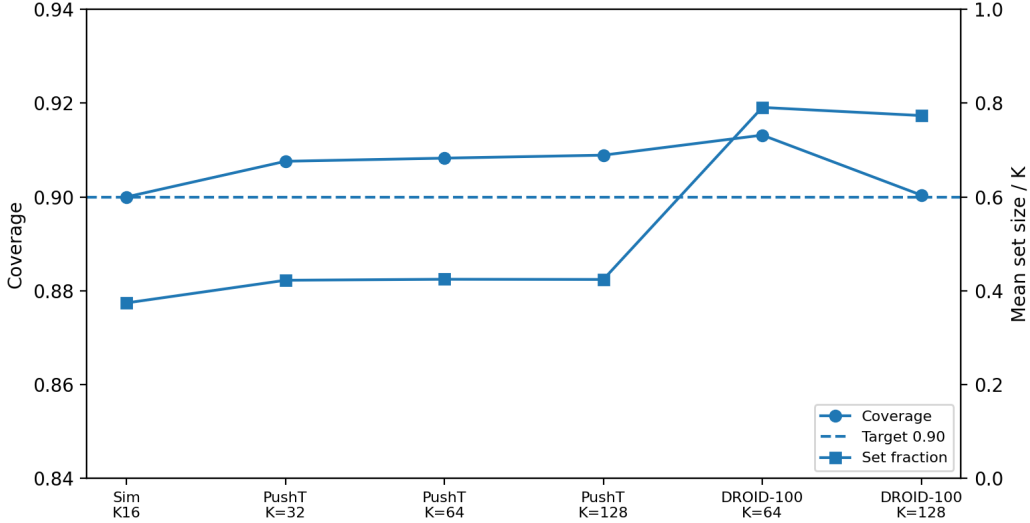


Figure 1: CoRAS coverage and normalized set size across tracks at $\alpha = 0.10$. The dashed line is the target coverage.

dataset but also exposes the limits of the current action-token predictor through large sets. AutoEval logs demonstrate compatibility with online real-robot evaluation traces but do not provide stable failure-prediction evidence at the available sample size.

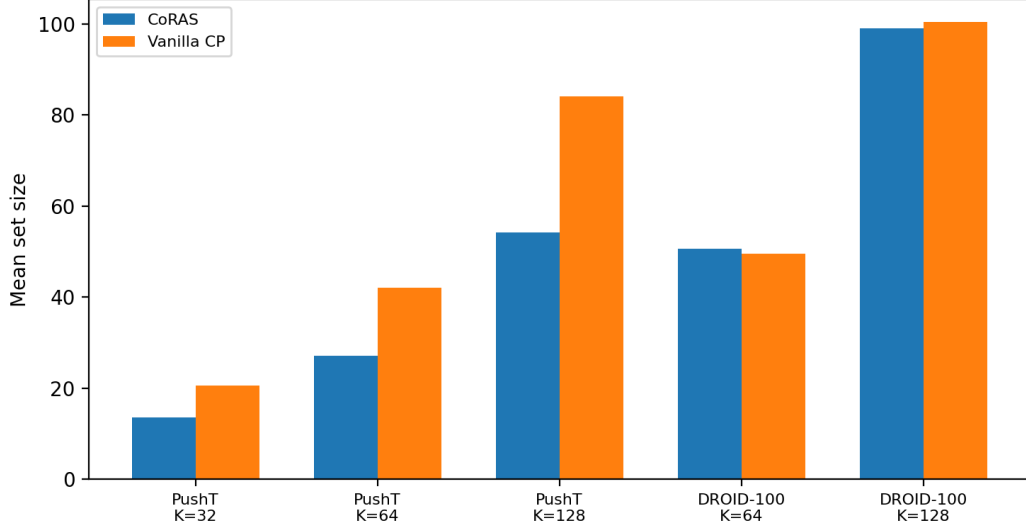


Figure 2: Mean set size for CoRAS and vanilla conformal prediction on real-data visual-shift tracks. CoRAS substantially reduces set size on PushT; on DROID-100, the action-token predictor remains too ambiguous for large reductions.

Table 9: Synthetic CoRAS sweep over error level and calibration fraction.

α	Calib. frac.	Runs	Coverage	$ \Delta $	Mean set	Top-1	Fail-to-abstain
0.05	0.10	5	0.955 ± 0.011	0.007 ± 0.010	9.479 ± 1.508	0.591 ± 0.033	0.000 ± 0.000
0.05	0.25	5	0.960 ± 0.011	0.012 ± 0.008	9.545 ± 1.213	0.608 ± 0.017	0.000 ± 0.001
0.05	0.40	5	0.950 ± 0.012	0.010 ± 0.005	8.997 ± 1.558	0.624 ± 0.039	0.001 ± 0.002
0.10	0.10	5	0.905 ± 0.027	0.022 ± 0.013	6.300 ± 1.763	0.591 ± 0.033	0.005 ± 0.006
0.10	0.25	5	0.909 ± 0.023	0.016 ± 0.018	6.284 ± 1.458	0.608 ± 0.017	0.004 ± 0.004
0.10	0.40	5	0.900 ± 0.019	0.013 ± 0.012	5.986 ± 1.211	0.624 ± 0.039	0.009 ± 0.008
0.20	0.10	5	0.803 ± 0.026	0.019 ± 0.015	3.111 ± 0.935	0.591 ± 0.033	0.043 ± 0.024
0.20	0.25	5	0.801 ± 0.026	0.019 ± 0.015	3.106 ± 0.881	0.608 ± 0.017	0.045 ± 0.025
0.20	0.40	5	0.805 ± 0.027	0.021 ± 0.014	3.050 ± 0.749	0.624 ± 0.039	0.041 ± 0.028

The table shows that decreasing α increases set size and coverage, while larger calibration fractions generally stabilize the empirical coverage gap.

Table 10: PushT action-codebook sweep.

K	Method	Coverage	Mean set	Set/K	Top-1	Action L2 gain
32	Top-1 singleton	0.145 ± 0.010	1.000 ± 0.000	0.031	0.145 ± 0.010	0.000 ± 0.000
32	Vanilla CP	0.914 ± 0.008	20.639 ± 0.440	0.645	0.145 ± 0.010	97.949 ± 5.383
32	Calibrated top-k	0.916 ± 0.010	22.556 ± 0.726	0.705	0.145 ± 0.010	98.365 ± 5.516
32	CoRAS	0.908 ± 0.010	13.517 ± 0.473	0.422	0.242 ± 0.012	68.852 ± 3.754
32	CoRAS-APS	0.906 ± 0.008	13.714 ± 0.579	0.429	0.242 ± 0.012	69.058 ± 3.839
64	Top-1 singleton	0.089 ± 0.008	1.000 ± 0.000	0.016	0.089 ± 0.008	0.000 ± 0.000
64	Vanilla CP	0.910 ± 0.016	42.036 ± 2.137	0.657	0.089 ± 0.008	107.211 ± 7.604
64	Calibrated top-k	0.913 ± 0.016	44.778 ± 2.224	0.700	0.089 ± 0.008	107.765 ± 7.410
64	CoRAS	0.908 ± 0.010	27.169 ± 1.029	0.425	0.155 ± 0.014	76.827 ± 3.729
64	CoRAS-APS	0.909 ± 0.010	27.308 ± 1.232	0.427	0.155 ± 0.014	76.973 ± 3.829
128	Top-1 singleton	0.053 ± 0.007	1.000 ± 0.000	0.008	0.053 ± 0.007	0.000 ± 0.000
128	Vanilla CP	0.913 ± 0.011	84.113 ± 3.336	0.657	0.053 ± 0.007	115.754 ± 6.298
128	Calibrated top-k	0.914 ± 0.010	90.556 ± 2.351	0.707	0.053 ± 0.007	116.000 ± 6.252
128	CoRAS	0.909 ± 0.010	54.285 ± 1.948	0.424	0.103 ± 0.008	83.003 ± 3.781
128	CoRAS-APS	0.909 ± 0.009	54.629 ± 2.021	0.427	0.103 ± 0.008	83.102 ± 3.780

Action L2 gain is the reduction from the top-1 action error to the best action contained in the prediction set, measured in the original PushT action coordinates.

Table 11: DROID-100 action-codebook sweep.

K	Method	Coverage	Mean set	Set/K	Top-1	Action L2 gain
64	Top-1 singleton	0.072 ± 0.020	1.000 ± 0.000	0.016	0.072 ± 0.020	0.000 ± 0.000
64	Vanilla CP	0.902 ± 0.043	49.533 ± 3.400	0.774	0.072 ± 0.020	0.464 ± 0.030
64	Calibrated top-k	0.901 ± 0.029	48.667 ± 2.345	0.760	0.072 ± 0.020	0.466 ± 0.029
64	CoRAS	0.913 ± 0.036	50.608 ± 3.360	0.791	0.060 ± 0.024	0.476 ± 0.032
64	CoRAS-APS	0.912 ± 0.034	50.508 ± 3.061	0.789	0.060 ± 0.024	0.476 ± 0.032
128	Top-1 singleton	0.064 ± 0.020	1.000 ± 0.000	0.008	0.064 ± 0.020	0.000 ± 0.000
128	Vanilla CP	0.905 ± 0.041	100.393 ± 6.545	0.784	0.064 ± 0.020	0.487 ± 0.024
128	Calibrated top-k	0.909 ± 0.032	100.778 ± 4.842	0.787	0.064 ± 0.020	0.489 ± 0.025
128	CoRAS	0.900 ± 0.041	98.994 ± 5.746	0.773	0.071 ± 0.018	0.436 ± 0.043
128	CoRAS-APS	0.898 ± 0.043	98.406 ± 5.839	0.769	0.071 ± 0.018	0.435 ± 0.043

The DROID sets occupy a larger fraction of the codebook than PushT, indicating that the current action-token predictor remains ambiguous on this more diverse real-robot dataset.

Table 12: PushT K=64 per-domain results at $\alpha = 0.10$.

Domain	Method	Coverage	Mean set	Top-1	Fail-to-abstain
shift_blur_noise	Top-1 singleton	0.069 $\pm$ 0.012	1.000 $\pm$ 0.000	0.069 $\pm$ 0.012	0.931 $\pm$ 0.012
shift_blur_noise	Vanilla CP	0.912 $\pm$ 0.030	46.102 $\pm$ 1.442	0.069 $\pm$ 0.012	0.000 $\pm$ 0.000
shift_blur_noise	Calibrated top-k	0.882 $\pm$ 0.025	44.778 $\pm$ 2.224	0.069 $\pm$ 0.012	0.000 $\pm$ 0.000
shift_blur_noise	CoRAS	0.910 $\pm$ 0.020	28.506 $\pm$ 1.046	0.150 $\pm$ 0.018	0.000 $\pm$ 0.000
shift_blur_noise	CoRAS-APS	0.913 $\pm$ 0.020	28.970 $\pm$ 0.868	0.150 $\pm$ 0.018	0.000 $\pm$ 0.000
shift_blur_noise	Mondrian CP	0.870 $\pm$ 0.047	24.192 $\pm$ 3.609	0.150 $\pm$ 0.018	0.000 $\pm$ 0.000
shift_camera_crop	Top-1 singleton	0.126 $\pm$ 0.012	1.000 $\pm$ 0.000	0.126 $\pm$ 0.012	0.874 $\pm$ 0.012
shift_camera_crop	Vanilla CP	0.969 $\pm$ 0.012	40.545 $\pm$ 1.767	0.126 $\pm$ 0.012	0.000 $\pm$ 0.000
shift_camera_crop	Calibrated top-k	0.964 $\pm$ 0.012	44.778 $\pm$ 2.224	0.126 $\pm$ 0.012	0.000 $\pm$ 0.000
shift_camera_crop	CoRAS	0.938 $\pm$ 0.012	22.299 $\pm$ 0.957	0.186 $\pm$ 0.023	0.000 $\pm$ 0.000
shift_camera_crop	CoRAS-APS	0.933 $\pm$ 0.011	21.849 $\pm$ 1.255	0.186 $\pm$ 0.023	0.000 $\pm$ 0.001
shift_camera_crop	Mondrian CP	0.909 $\pm$ 0.018	18.832 $\pm$ 1.917	0.186 $\pm$ 0.023	0.000 $\pm$ 0.000
shift_lighting	Top-1 singleton	0.083 $\pm$ 0.012	1.000 $\pm$ 0.000	0.083 $\pm$ 0.012	0.917 $\pm$ 0.012
shift_lighting	Vanilla CP	0.848 $\pm$ 0.032	40.459 $\pm$ 2.718	0.083 $\pm$ 0.012	0.000 $\pm$ 0.000
shift_lighting	Calibrated top-k	0.885 $\pm$ 0.026	44.778 $\pm$ 2.224	0.083 $\pm$ 0.012	0.000 $\pm$ 0.000
shift_lighting	CoRAS	0.905 $\pm$ 0.022	29.312 $\pm$ 1.643	0.164 $\pm$ 0.026	0.000 $\pm$ 0.000
shift_lighting	CoRAS-APS	0.908 $\pm$ 0.018	29.657 $\pm$ 2.068	0.164 $\pm$ 0.026	0.000 $\pm$ 0.001
shift_lighting	Mondrian CP	0.916 $\pm$ 0.022	31.009 $\pm$ 4.639	0.164 $\pm$ 0.026	0.000 $\pm$ 0.000
shift_occlusion	Top-1 singleton	0.083 $\pm$ 0.018	1.000 $\pm$ 0.000	0.083 $\pm$ 0.018	0.917 $\pm$ 0.018
shift_occlusion	Vanilla CP	0.916 $\pm$ 0.028	40.444 $\pm$ 2.389	0.083 $\pm$ 0.018	0.000 $\pm$ 0.000
shift_occlusion	Calibrated top-k	0.928 $\pm$ 0.021	44.778 $\pm$ 2.224	0.083 $\pm$ 0.018	0.000 $\pm$ 0.000
shift_occlusion	CoRAS	0.874 $\pm$ 0.046	28.916 $\pm$ 1.087	0.113 $\pm$ 0.024	0.000 $\pm$ 0.000
shift_occlusion	CoRAS-APS	0.874 $\pm$ 0.042	29.200 $\pm$ 1.096	0.113 $\pm$ 0.024	0.000 $\pm$ 0.000
shift_occlusion	Mondrian CP	0.914 $\pm$ 0.037	33.672 $\pm$ 3.272	0.113 $\pm$ 0.024	0.000 $\pm$ 0.000

Per-domain metrics show where calibration is hardest. Coverage is marginal within each visual-shift domain and is not guaranteed conditionally by the split-conformal theorem.

Table 13: DROID-100 K=64 per-domain results at $\alpha = 0.10$.

Domain	Method	Coverage	Mean set	Top-1	Fail-to-abstain
shift_blur_noise	Top-1 singleton	0.133 $\pm$ 0.104	1.000 $\pm$ 0.000	0.133 $\pm$ 0.104	0.867 $\pm$ 0.104
shift_blur_noise	Vanilla CP	0.910 $\pm$ 0.107	49.016 $\pm$ 5.754	0.133 $\pm$ 0.104	0.000 $\pm$ 0.000
shift_blur_noise	Calibrated top-k	0.931 $\pm$ 0.061	48.667 $\pm$ 2.345	0.133 $\pm$ 0.104	0.000 $\pm$ 0.000
shift_blur_noise	CoRAS	0.913 $\pm$ 0.078	51.362 $\pm$ 5.274	0.091 $\pm$ 0.068	0.000 $\pm$ 0.000
shift_blur_noise	CoRAS-APS	0.910 $\pm$ 0.076	51.087 $\pm$ 4.594	0.091 $\pm$ 0.068	0.000 $\pm$ 0.000
shift_blur_noise	Mondrian CP	0.912 $\pm$ 0.111	52.483 $\pm$ 7.559	0.091 $\pm$ 0.068	0.000 $\pm$ 0.000
shift_camera_crop	Top-1 singleton	0.083 $\pm$ 0.029	1.000 $\pm$ 0.000	0.083 $\pm$ 0.029	0.917 $\pm$ 0.029
shift_camera_crop	Vanilla CP	0.890 $\pm$ 0.049	50.152 $\pm$ 5.584	0.083 $\pm$ 0.029	0.000 $\pm$ 0.000
shift_camera_crop	Calibrated top-k	0.871 $\pm$ 0.030	48.667 $\pm$ 2.345	0.083 $\pm$ 0.029	0.000 $\pm$ 0.000
shift_camera_crop	CoRAS	0.897 $\pm$ 0.032	51.453 $\pm$ 2.072	0.050 $\pm$ 0.016	0.000 $\pm$ 0.000
shift_camera_crop	CoRAS-APS	0.895 $\pm$ 0.031	51.219 $\pm$ 2.015	0.050 $\pm$ 0.016	0.000 $\pm$ 0.000
shift_camera_crop	Mondrian CP	0.907 $\pm$ 0.050	52.760 $\pm$ 3.476	0.050 $\pm$ 0.016	0.000 $\pm$ 0.000
shift_lighting	Top-1 singleton	0.044 $\pm$ 0.024	1.000 $\pm$ 0.000	0.044 $\pm$ 0.024	0.956 $\pm$ 0.024
shift_lighting	Vanilla CP	0.913 $\pm$ 0.038	49.268 $\pm$ 2.265	0.044 $\pm$ 0.024	0.000 $\pm$ 0.000
shift_lighting	Calibrated top-k	0.914 $\pm$ 0.031	48.667 $\pm$ 2.345	0.044 $\pm$ 0.024	0.000 $\pm$ 0.000
shift_lighting	CoRAS	0.933 $\pm$ 0.048	52.391 $\pm$ 2.726	0.038 $\pm$ 0.018	0.000 $\pm$ 0.000
shift_lighting	CoRAS-APS	0.932 $\pm$ 0.048	52.180 $\pm$ 2.628	0.038 $\pm$ 0.018	0.000 $\pm$ 0.000
shift_lighting	Mondrian CP	0.946 $\pm$ 0.045	54.347 $\pm$ 3.674	0.038 $\pm$ 0.018	0.000 $\pm$ 0.000
shift_occlusion	Top-1 singleton	0.069 $\pm$ 0.054	1.000 $\pm$ 0.000	0.069 $\pm$ 0.054	0.931 $\pm$ 0.054
shift_occlusion	Vanilla CP	0.901 $\pm$ 0.053	48.359 $\pm$ 3.003	0.069 $\pm$ 0.054	0.000 $\pm$ 0.000
shift_occlusion	Calibrated top-k	0.907 $\pm$ 0.051	49.000 $\pm$ 2.268	0.069 $\pm$ 0.054	0.000 $\pm$ 0.000
shift_occlusion	CoRAS	0.937 $\pm$ 0.047	49.699 $\pm$ 4.999	0.073 $\pm$ 0.065	0.000 $\pm$ 0.000
shift_occlusion	CoRAS-APS	0.938 $\pm$ 0.045	49.627 $\pm$ 4.325	0.073 $\pm$ 0.065	0.000 $\pm$ 0.000
shift_occlusion	Mondrian CP	0.890 $\pm$ 0.078	44.888 $\pm$ 6.208	0.073 $\pm$ 0.065	0.000 $\pm$ 0.000

Per-domain metrics show where calibration is hardest. Coverage is marginal within each visual-shift domain and is not guaranteed conditionally by the split-conformal theorem.

Table 14: PushT K=64 continuous-action geometry at $\alpha = 0.10$.

Method	Top-1 L2	Oracle-in-set L2	L2 gain	Set diameter
Top-1 singleton	128.896 ± 7.219	128.896 ± 7.219	0.000 ± 0.000	0.000 ± 0.000
Prompt top-1	98.253 ± 3.873	98.253 ± 3.873	0.000 ± 0.000	0.000 ± 0.000
Vanilla CP	128.896 ± 7.219	21.685 ± 0.816	107.211 ± 7.604	448.086 ± 13.134
Calibrated top-k	128.896 ± 7.219	21.131 ± 0.688	107.765 ± 7.410	471.041 ± 10.739
CoRAS	98.253 ± 3.873	21.426 ± 0.484	76.827 ± 3.729	368.319 ± 11.092
CoRAS-APS	98.253 ± 3.873	21.280 ± 0.477	76.973 ± 3.829	366.546 ± 12.730

Oracle-in-set L2 reports the distance from the demonstrated continuous action to the nearest codebook center contained in the prediction set. It is not an executable oracle; it measures whether the set contains geometrically relevant candidates.

Table 15: DROID-100 K=64 continuous-action geometry at $\alpha = 0.10$.

Method	Top-1 L2	Oracle-in-set L2	L2 gain	Set diameter
Top-1 singleton	0.805 ± 0.043	0.805 ± 0.043	0.000 ± 0.000	0.000 ± 0.000
Prompt top-1	0.814 ± 0.047	0.814 ± 0.047	0.000 ± 0.000	0.000 ± 0.000
Vanilla CP	0.805 ± 0.043	0.340 ± 0.027	0.464 ± 0.030	2.010 ± 0.042
Calibrated top-k	0.805 ± 0.043	0.339 ± 0.025	0.466 ± 0.029	2.023 ± 0.035
CoRAS	0.814 ± 0.047	0.338 ± 0.025	0.476 ± 0.032	2.036 ± 0.046
CoRAS-APS	0.814 ± 0.047	0.338 ± 0.025	0.476 ± 0.032	2.037 ± 0.046

Oracle-in-set L2 reports the distance from the demonstrated continuous action to the nearest codebook center contained in the prediction set. It is not an executable oracle; it measures whether the set contains geometrically relevant candidates.

Table 16: PushT K=64 episode/block diagnostics at $\alpha = 0.10$.

Method	Episode mean cov.	All-step episode rate	Block boot lo95	Block boot hi95
Top-1 singleton	0.091 ± 0.009	0.000 ± 0.000	0.071 ± 0.008	0.108 ± 0.010
Prompt top-1	0.156 ± 0.016	0.000 ± 0.000	0.132 ± 0.013	0.178 ± 0.017
Vanilla CP	0.911 ± 0.014	0.028 ± 0.036	0.881 ± 0.021	0.936 ± 0.013
Calibrated top-k	0.916 ± 0.014	0.033 ± 0.035	0.887 ± 0.019	0.937 ± 0.012
CoRAS	0.907 ± 0.012	0.017 ± 0.035	0.884 ± 0.015	0.931 ± 0.008
CoRAS-APS	0.907 ± 0.011	0.017 ± 0.035	0.885 ± 0.014	0.930 ± 0.008
Mondrian CP	0.900 ± 0.021	0.011 ± 0.022	0.876 ± 0.028	0.921 ± 0.018

Block bootstrap intervals are computed over episode blocks, reducing the influence of frame-level temporal correlation.

Table 17: DROID-100 K=64 episode/block diagnostics at $\alpha = 0.10$.

Method	Episode mean cov.	All-step episode rate	Block boot lo95	Block boot hi95
Top-1 singleton	0.083 ± 0.027	0.000 ± 0.000	0.040 ± 0.008	0.122 ± 0.042
Prompt top-1	0.061 ± 0.027	0.000 ± 0.000	0.038 ± 0.013	0.085 ± 0.040
Vanilla CP	0.893 ± 0.050	0.000 ± 0.000	0.850 ± 0.078	0.943 ± 0.022
Calibrated top-k	0.900 ± 0.037	0.012 ± 0.037	0.862 ± 0.035	0.938 ± 0.023
CoRAS	0.915 ± 0.045	0.037 ± 0.079	0.870 ± 0.050	0.954 ± 0.028
CoRAS-APS	0.913 ± 0.043	0.049 ± 0.081	0.868 ± 0.048	0.953 ± 0.026
Mondrian CP	0.911 ± 0.045	0.086 ± 0.121	0.850 ± 0.068	0.957 ± 0.026

Block bootstrap intervals are computed over episode blocks, reducing the influence of frame-level temporal correlation.

Table 18: AutoEval online-log diagnostic results.

Subset	Method	Coverage	Mean set	Top-1	Oracle L2	Episode mean cov.
K=32	Top-1 singleton	0.136	1.000	0.136	0.259	0.115
K=32	Prompt top-1	0.045	1.000	0.045	0.314	0.038
K=32	Vanilla CP	1.000	27.727	0.136	0.036	1.000
K=32	Calibrated top-k	1.000	26.000	0.136	0.036	1.000
K=32	CoRAS	0.864	13.364	0.045	0.050	0.868
K=32	CoRAS-APS	0.909	13.227	0.045	0.050	0.923
K=64	Top-1 singleton	0.096	1.000	0.096	0.173	0.096
K=64	Prompt top-1	0.048	1.000	0.048	0.229	0.046
K=64	Vanilla CP	0.885	49.173	0.096	0.041	0.876
K=64	Calibrated top-k	0.875	49.000	0.096	0.041	0.871
K=64	CoRAS	0.885	44.317	0.048	0.043	0.882
K=64	CoRAS-APS	0.894	45.327	0.048	0.042	0.890

AutoEval results are auxiliary. The K=32 run is a bounded small-scale aggregate; the K=64 table reports the completed diagnostic subset with raw action labels. Failure-prediction AUCs on the K=64 subset are near chance and are therefore not used as main evidence.

Table 19: AutoEval K=64 failure-risk diagnostics for CoRAS.

Run	Frames	Episodes	Frame AUC	Episode mean AUC	Episode max AUC	Unc. success	Unc. failure
seed0 0.10	104	5	0.502	0.500	0.333	44.250	44.350
seed0 0.25	154	8	0.435	0.250	0.167	41.360	39.220
seed0 0.40	104	5	0.473	0.000	0.667	39.060	38.560
seed1 0.10	111	6	0.469	0.500	0.375	34.390	33.630
seed1 0.25	104	5	0.454	0.000	0.667	39.080	38.500
seed1 0.40	147	7	0.579	0.900	0.200	37.390	39.830
seed2 0.10	154	8	0.564	0.833	0.458	41.750	42.990
seed2 0.25	104	5	0.471	0.167	0.333	46.690	45.780

Set size was evaluated as a failure-risk score. The small episode counts and unstable AUCs indicate that these online logs establish applicability of the parser and calibration pipeline, not reliable failure prediction.